UPAR Challenge 2027: Track 2

Attribute-Based Pedestrian Retrieval



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1. Abstract

This project addresses attribute-based pedestrian retrieval for UPAR Challenge 2027 Track 2. Given a query specifying 40 binary pedestrian attributes, the system ranks gallery images by how well the people depicted match that attribute description. The experiments compared a ResNet-50 baseline, ConvNeXt-Tiny attribute prediction, and two retrieval-oriented systems. The final submission candidate combines the V3 VisualQuery and CLEAN Hybrid systems using a fixed 65:35 score-level ensemble. On the complete public validation set of 3,462 queries, the ensemble achieved 69.63% mADM. This is a public validation result, not a verified hidden-test or official leaderboard score.

2. Problem Statement and Objectives

The goal is to rank gallery images using a 40-attribute binary query. The organizer's ground truth groups images by their complete 40-bit attribute descriptions; the learned model estimates attribute compatibility and ranks images by distance, even when its predictions are imperfect.

The main objectives were to establish a baseline, compare visual backbones and retrieval scoring approaches, evaluate systems on a consistent public validation set, and package the strongest tested approach for Codabench submission.

3. Dataset and Evaluation

The research record describes public training data comprising 97,669 images from Market1501, PA-100K, and PETA. The public validation evaluation uses 3,462 queries and a gallery of 33,407 images. Each query ranks the full gallery. The primary reported retrieval metric is mADM, which evaluates ranking quality with attribute agreement taken into account. mAP and Rank-1 were measured for some individual experiments; Rank-5, Rank-10, and mINP were not consistently available across all runs.

All reported headline results below are public validation measurements. The 600-query cohort used in some earlier experiments is reported separately and should not be compared directly with full-validation scores.

4. Methodology

4.1 Baseline Models

A ResNet-50 baseline established a reference point. ConvNeXt-Tiny was then evaluated as an alternative visual backbone for predicting the 40 binary attributes. The ConvNeXt-Tiny baseline improved over ResNet-50 on the complete public validation set.

4.2 Retrieval-Oriented Systems

The V3 VisualQuery system adds image and query retrieval heads over frozen ConvNeXt-Tiny baseline features. It is a retrieval-oriented model built on the baseline features, rather than a separate independent backbone. The CLEAN Hybrid system provides another retrieval scoring approach, combining attribute-based and cosine-distance signals.

4.3 Final Ensemble

The final candidate combines V3 VisualQuery at 65% and CLEAN Hybrid at 35%. Before combining scores, each system's scores are normalized per query across the gallery using z-normalization. V3 internally combines Bernoulli negative log-likelihood and scaled cosine distance with equal weights. The Hybrid scorer combines L1 attribute distance at 60% and cosine distance at 40%. The ensemble is a fixed score-level combination, not a jointly trained end-to-end model.

5. Results

System	Scoring / Configuration	mADM	Evaluation
ResNet-50 baseline	Uniform NLL	58.53%	3,462 queries
ConvNeXt-Tiny baseline	Uniform NLL	63.13%	3,462 queries
V3 VisualQuery	Retrieval head over frozen ConvNeXt-Tiny features	66.56%	3,462 queries
CLEAN Hybrid	L1 and cosine scoring	67.26%	3,462 queries
Final ensemble	V3 65% + Hybrid 35%, per-query gallery z-normalization	69.63%	3,462 queries
Final ensemble, earlier cohort	Same ensemble	71.22%	600-query cohort only

The best full public-validation result in the research record is 69.63% mADM. The Codabench screenshot records submission 1001454 as Finished with a displayed score of 0.5361, compared with 0.4826 for earlier submission 1000643. The local ensemble score of 0.6963 is from public validation and must not be presented as the official score. Because the displayed Codabench score differs, the precise evaluation phase, metric and data split should be checked before interpreting this gap.

6. Challenges and Error Analysis

A separate saved attribute-prediction audit found an average of 2.3766 incorrect binary attributes per image at a probability threshold of 0.5. Only 20.18% of gallery images had all 40 attributes predicted correctly. Bags, normal glasses, backpacks, and lower-body clothing colours were among the difficult attributes.

Key challenges included domain shift between datasets, fine-grained visual distinctions, attribute imbalance, public-validation model-selection bias, compute constraints, dataset download issues in Colab, and TorchScript runtime compatibility. Repeated tuning against public validation can lead to optimistic estimates of generalization. A completed scored Codabench run demonstrates successful execution on that platform; separate reload, runtime and reproducibility checks remain useful.

7. Possible Improvements

The following are proposed next steps, not completed or verified improvements:

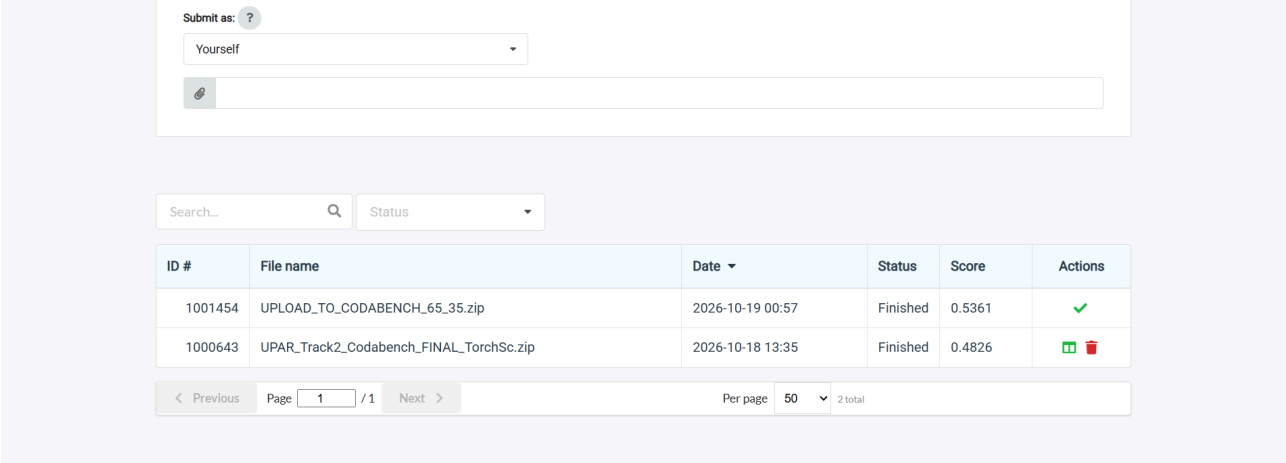
- Improve attribute-specific learning through carefully validated class weighting, focal loss, or balanced sampling, especially for rare and visually ambiguous attributes.
- Add stronger data augmentation and domain generalization experiments while preserving realistic pedestrian appearance and attribute labels.
- Evaluate higher-capacity backbones or complementary pretrained features only when compute permits, and compare them on the same full validation protocol.
- Tune ensemble weights and score normalization on a separate development split or through cross-validation to reduce overfitting to the public validation set.
- Investigate attribute-specific calibration and uncertainty-aware scoring, but retain these methods only if they improve the official metric on a held-out evaluation.
- Run an end-to-end reproducibility check of the final archive, including model loading, dependency versions, input preprocessing, output format, runtime, and memory use in an environment close to Codabench.
- Report all required metrics, including mAP, Rank-1, Rank-5, Rank-10, and mINP, from the same evaluation protocol where available.

8. Submission Artifact and Codabench Evidence

The prepared submission candidate is UPLOAD_TO_CODABENCH_65_35.zip, approximately 212 MB according to the research record. It contains run.py, metadata.yaml, configuration, and four TorchScript files: v3_image.ts, v3_query.ts, hybrid_image.ts, and hybrid_query.ts. These files support two retrieval systems used by the ensemble; they do not represent four independent backbones.

The larger research collection holds training checkpoints and analysis material; it is not the intended Codabench submission. The competition ZIP should be submitted separately from the report and other assignment deliverables.

8.1 Codabench Submission Confirmation



41	hamzasheikh2605	2026-10-19 00:57	1001454	1001454_VID-Docker-GPU	**0.5361**	0.2604	0.2859	0.4691	0.5668	0.191

9. Conclusion

The experiments show a consistent improvement from the ResNet-50 baseline to ConvNeXt-Tiny and then to the retrieval-oriented systems. The fixed V3 and CLEAN Hybrid ensemble achieved 69.63% mADM on the full public validation set and is the strongest tested candidate documented in the research record. Future work should focus on more robust held-out validation, difficult attribute categories, reproducibility, and investigating the difference between public-validation and displayed Codabench scores.

10. References

- UPAR Challenge 2027, Track 2 competition page: https://www.codabench.org/competitions/18196/
- UPAR Challenge 2027 organizer repository: https://github.com/speckean/UPAR-Challenge-2027
- Project repository: https://github.com/hamza26051/cv-assignment

• Specker, A., Cormier, M., and Beyerer, J. (2023). “UPAR: Unified Pedestrian Attribute Recognition and Person Retrieval.” Proceedings of WACV.